

Algebraic Geometry and Coding Theory

Lecture 3

September 24

Main theme. Reed–Solomon codes evaluate polynomials on the projective line. Replacing the line by a curve gives more possible evaluation points, while Riemann–Roch controls the dimension lost to genus. Families with many rational points relative to genus yield the TVZ bound, which improves the Gilbert–Varshamov bound. We then sketch the geometry behind Riemann–Roch and two approaches to the Hasse–Weil point bound.

Throughout, q is a prime power. We write $m = \deg D$ for a divisor degree and $[n, r, d]_q$ for code parameters, keeping the minimum distance d distinct from the pole bound m . The code dimension r is the parameter often denoted by k .

Contents

1	Reed–Solomon codes as functions on the projective line	2
1.1	Polynomial degree is a pole bound	2
1.2	The obstruction is the length	3
2	Evaluation codes on an algebraic curve	3
2.1	Definition and the kernel of evaluation	3
2.2	The minimum-distance estimate	4
2.3	Riemann–Roch supplies the dimension	4
3	Why the ratio of rational points to genus matters	5
3.1	Rates, distances, and a geometric optimization problem	5
3.2	From a family of curves to an asymptotic coding bound	6
4	Point-counting theorems and the TVZ bound	7
5	Comparing the TVZ line with the GV curve	8
5.1	The calculation behind the graphs	8
6	A geometric explanation of Riemann–Roch	12
6.1	Local function spaces, cohomology, and genus	12
6.2	Adding a point increases Euler characteristic by one	13
6.3	Serre duality identifies the correction term	13
6.4	The large-degree formula and the general field	14
7	Hasse–Weil from intersections on a surface	15
7.1	Rational points as fixed points of Frobenius	15
7.2	The intersection table	15
7.3	Project away from the fibers and use Hodge index	16
8	Hasse–Weil from the zeta function and function-field RH	17
8.1	Effective divisors and the zeta function	17
8.2	The polynomial numerator and its first coefficient	18
8.3	Weil’s Riemann hypothesis and the point bound	19

1 Reed–Solomon codes as functions on the projective line

The basic mechanism of Reed–Solomon codes is worth recalling before generalizing it. Choose distinct elements $a_1, \dots, a_n \in \mathbb{F}_q$ and an integer $0 \leq m < n$. Evaluate all polynomials of degree at most m :

$$\text{ev} : \{f(t) \in \mathbb{F}_q[t] : \deg f \leq m\} \longrightarrow \mathbb{F}_q^n, \quad f \longmapsto (f(a_1), \dots, f(a_n)).$$

The domain has basis $1, t, \dots, t^m$, so it has dimension $m+1$. A nonzero polynomial of degree at most m cannot vanish at all n evaluation points. Thus evaluation is injective. Moreover, it vanishes at at most m of the chosen points, so every nonzero codeword has at least $n - m$ nonzero coordinates.

The resulting code has dimension $m+1$ and minimum distance at least $n - m$. Singleton gives the opposite inequality

$$d \leq n - (m+1) + 1 = n - m.$$

Therefore its exact parameters are

$$\boxed{[n, m+1, n-m]_q}. \quad (1)$$

This is the evaluation description from Lecture 1; compare also Morandi's discussion in [M, §5.1].

1.1 Polynomial degree is a pole bound

The projective line has rational points

$$\mathbb{P}^1(\mathbb{F}_q) = \{[a : 1] : a \in \mathbb{F}_q\} \cup \{\infty\}, \quad \infty = [1 : 0],$$

and function field $\mathbb{F}_q(\mathbb{P}^1) = \mathbb{F}_q(t)$. A local coordinate at infinity is $u = 1/t$. Consequently, a polynomial of degree j has a pole of order j at infinity and no other poles.

For the divisor $D = m\infty$, the definition from Lecture 2 becomes

$$L(m\infty) = \{f \in \mathbb{F}_q(t) : (f) + m\infty \geq 0\} \cup \{0\}.$$

A rational function with no poles on the affine line is a polynomial: in a reduced fraction, any nonconstant denominator has an irreducible factor giving a finite pole. The condition at infinity then bounds its degree. Hence

$$\boxed{L(m\infty) = \text{Span}_{\mathbb{F}_q}\{1, t, \dots, t^m\}}. \quad (2)$$

Thus Reed–Solomon is the evaluation of a Riemann–Roch space at the rational points $P_i = [a_i : 1]$, away from the chosen pole point.

Example 1.1. Over $\mathbb{F}_5$, take all five affine points and $D = 2\infty$. Evaluating the basis $1, t, t^2$ gives the generator matrix

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 4 & 4 & 1 \end{pmatrix}.$$

The parameters are $[5, 3, 3]_5$. The function $t(t-1)$ has two zeros among the evaluation points, so its codeword has weight three.

1.2 The obstruction is the length

The construction above requires $n \leq q$: there are only q available finite rational points. The projective extension from Lecture 1 reaches length $q + 1$, using a suitable extra coordinate at infinity, but even that length is bounded when the alphabet is fixed. It is not ordinary evaluation of a function at a pole.

Our objective is to keep $\mathbb{F}_q$ fixed and make the length tend to infinity. This suggests replacing $\mathbb{P}^1$ by other curves with more rational points. We must replace two polynomial facts at once: the dimension of the function space and the bound on the number of zeros.

2 Evaluation codes on an algebraic curve

Let $X/\mathbb{F}_q$ be a smooth projective geometrically integral curve of genus g . Here the curve need not be embedded in $\mathbb{P}^2$; it can be described by compatible affine charts in a larger projective space. Smoothness excludes singular points, projectivity includes the points needed to account for all poles, and geometric integrality says that the curve remains irreducible and reduced over $\overline{\mathbb{F}_q}$. We denote its rational-function field by $\mathbb{F}_q(X)$.

Recall the divisor notation

$$D = \sum_P n_P P, \quad \deg D = \sum_P n_P \deg P, \quad (f) = \sum_P \text{ord}_P(f) P.$$

The sum for a divisor has finite support. Over $\mathbb{F}_q$, a closed point of degree e is the Frobenius orbit of e geometric points discussed in Lecture 2. Its degree is also $[\kappa(P) : \mathbb{F}_q]$. A rational point has degree one.

For a divisor D , put

$$L(D) = \{f \in \mathbb{F}_q(X)^\times : (f) + D \geq 0\} \cup \{0\}, \quad \ell(D) = \dim_{\mathbb{F}_q} L(D).$$

The inequality means that $\text{ord}_P(f) \geq -n_P$ at every point. In particular, a point outside $\text{supp}(D)$ is not allowed to be a pole. We use the facts that $\deg(f) = 0$ and that an effective divisor has nonnegative degree. They immediately imply

$$\deg D < 0 \implies L(D) = 0. \quad (3)$$

Indeed, a nonzero $f \in L(D)$ would make $(f) + D$ effective of degree $\deg D < 0$.

2.1 Definition and the kernel of evaluation

Choose distinct rational points

$$\mathcal{P} = \{P_1, \dots, P_n\} \subset X(\mathbb{F}_q), \quad \text{supp}(D) \cap \mathcal{P} = \emptyset,$$

and let $E = P_1 + \dots + P_n$ be the evaluation divisor. Every $f \in L(D)$ is regular at P_i , and its value lies in $\kappa(P_i) = \mathbb{F}_q$. Thus

$$\text{ev}_{\mathcal{P}} : L(D) \longrightarrow \mathbb{F}_q^n, \quad f \longmapsto (f(P_1), \dots, f(P_n))$$

is a well-defined linear map.

Definition 2.1. The *algebraic geometry evaluation code*, or *geometric Goppa code*, associated to $(X, \mathcal{P}, D)$ is

$$C(X, \mathcal{P}, D) = \text{im}(\text{ev}_{\mathcal{P}}).$$

We use Walker's notation $C(X, \mathcal{P}, D)$. In Morandi's notation the same construction is $C_L(E, D)$: his first divisor records the evaluation points and his second controls the allowed functions. See [W, Chapter 6] and [M, Chapter 5].

Proposition 2.2. *The evaluation map has kernel $L(D - E)$. Hence*

$$\dim C(X, \mathcal{P}, D) = \ell(D) - \ell(D - E).$$

If $\deg D < n$, evaluation is injective and the dimension is $\ell(D)$.

Proof. Since the coefficient of P_i in D is zero, the condition $f(P_i) = 0$ is equivalent to $\text{ord}_{P_i}(f) \geq 1$. Requiring this at all evaluation points, while retaining the other bounds defining $L(D)$, is exactly

$$(f) + D - E \geq 0.$$

This proves the kernel formula. The dimension statement follows from rank-nullity. Finally,

$$\deg(D - E) = \deg D - n < 0$$

under the indicated hypothesis, so (3) gives $L(D - E) = 0$. $\square$

2.2 The minimum-distance estimate

Proposition 2.3. *If $C(X, \mathcal{P}, D)$ is nonzero, its minimum distance satisfies*

$$\boxed{d \geq n - \deg D.} \quad (4)$$

Proof. Let $\text{ev}_{\mathcal{P}}(f)$ be a nonzero codeword, and suppose that z of its coordinates vanish. The function f is nonzero, and $(f) + D$ is effective of degree $\deg D$. Each of the z zero coordinates contributes at least one to this degree, because its evaluation point has degree one and coefficient zero in D . Therefore

$$z \leq \deg D, \quad \text{wt}(\text{ev}_{\mathcal{P}}(f)) = n - z \geq n - \deg D.$$

Taking the minimum over nonzero codewords proves the claim. $\square$

Remark 2.4. The argument counts zeros using the *effective divisor* $(f) + D$. This formulation works even when D has negative coefficients. The number $n - \deg D$ is the *designed distance*; the actual minimum distance may be larger.

2.3 Riemann–Roch supplies the dimension

The form of Riemann–Roch stated in Lecture 2 says

$$\ell(D) \geq \deg D + 1 - g, \quad \ell(D) = \deg D + 1 - g \quad \text{if } \deg D > 2g - 2. \quad (5)$$

The genus is zero for $\mathbb{P}^1$ and one for a smooth plane cubic. Section 6 explains the geometric definition and the reason for the threshold $2g - 2$.

Theorem 2.5 (AG-code parameters). *Suppose $\mathcal{P}$ consists of n distinct rational points disjoint from $\text{supp}(D)$, and write $m = \deg D$. If $m < n$ and $L(D) \neq 0$, then $C(X, \mathcal{P}, D)$ has parameters $[n, r, d]_q$ with*

$$\boxed{r = \ell(D) \geq m + 1 - g, \quad d \geq n - m.} \quad (6)$$

In particular, if $0 \leq m < n$ and $m > 2g - 2$, its parameters are

$$\boxed{[n, m + 1 - g, d]_q, \quad d \geq n - m.} \quad (7)$$

Proof. Combine Propositions 2.2 and 2.3 with (5). In the final range, $m + 1 - g > 0$, so the code is nonzero. $\square$

This is Walker's Theorem 6.4 in its equality range, together with the general dimension inequality from Morandi's Corollary 5.3. If $f_1, \dots, f_r$ is a basis of $L(D)$ and $m < n$, then a generator matrix has entries $f_j(P_i)$: its rows are the evaluations of that basis.

Compared with Reed–Solomon, the guaranteed distance can fall short of the MDS value. In the equality range, Singleton and (4) give

$$n - m \leq d \leq n - r + 1 = n - m + g.$$

More generally, in the nonzero injective range,

$$\boxed{0 \leq n + 1 - r - d \leq g.} \quad (8)$$

Thus genus bounds the possible Singleton defect; it need not equal the defect. The advantage is not that every individual AG code is better than Reed–Solomon. It is that a suitable curve may supply far more evaluation points over the same field.

3 Why the ratio of rational points to genus matters

3.1 Rates, distances, and a geometric optimization problem

For a linear $[n, r, d]_q$ code, recall its information rate and relative minimum distance:

$$R = \frac{r}{n}, \quad \delta = \frac{d}{n}.$$

For a not necessarily linear code, replace r by the base- q logarithm of its size. As in Walker, §2.2, let $A_q(n, d)$ be the largest size of a code of length n with minimum distance at least d , and define

$$\alpha_q(\delta) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log_q A_q(n, \lfloor \delta n \rfloor), \quad 0 \leq \delta \leq 1.$$

We use $A_q(n, 0) = q^n$. The function α_q measures the best asymptotic rate available with a prescribed relative-distance lower bound.

For an AG code, (6) gives

$$R + \delta \geq \frac{m + 1 - g}{n} + \frac{n - m}{n} = 1 + \frac{1}{n} - \frac{g}{n}. \quad (9)$$

Increasing m helps the dimension but weakens the distance guarantee. Increasing the number of evaluation points, without increasing the genus by the same proportion, helps the combined estimate. The relevant penalty is g/n , not the genus or the number of points separately.

Let $N = \#X(\mathbb{F}_q)$. A convenient one-point construction reserves $O \in X(\mathbb{F}_q)$ for poles and evaluates at the other $N - 1$ points. Then

$$D = mO, \quad n = N - 1.$$

This motivates the definitions from Walker's Chapter 7.

Definition 3.1. For a fixed prime power q , let

$$N_q(g) = \max\{\#X(\mathbb{F}_q) : X/\mathbb{F}_q \text{ smooth projective geometrically integral of genus } g\}.$$

The *Ihara constant* is

$$A(q) = \limsup_{g \rightarrow \infty} \frac{N_q(g)}{g}. \quad (10)$$

The field is fixed in this definition. Passing from $\mathbb{F}_q$ to larger and larger fields changes the alphabet and does not solve the fixed-alphabet problem. We also cannot insist that all the curves be smooth plane curves: a curve embedded in $\mathbb{P}_{\mathbb{F}_q}^2$ has at most $q^2 + q + 1$ rational points. Arbitrarily long evaluation codes require a more general family of smooth projective curves.

3.2 From a family of curves to an asymptotic coding bound

The sum inequality (9) suggests the answer, but to obtain a bound for *each chosen* δ we must select the divisors accordingly. The following makes that choice explicit.

Proposition 3.2. Suppose $X_i/\mathbb{F}_q$ is a sequence of curves with

$$g_i = g(X_i) \rightarrow \infty, \quad N_i = \#X_i(\mathbb{F}_q), \quad \frac{N_i}{g_i} \rightarrow A > 0.$$

Then, for $0 \leq \delta \leq 1$,

$$\boxed{\alpha_q(\delta) \geq \max\left\{0, 1 - \delta - \frac{1}{A}\right\}}. \quad (11)$$

Proof. Fix $0 < \delta < 1$. Since $A > 0$ and $g_i \rightarrow \infty$, we have $N_i \rightarrow \infty$. For all sufficiently large i , choose $O_i \in X_i(\mathbb{F}_q)$ and evaluate at every other rational point. Put

$$n_i = N_i - 1, \quad m_i = \lfloor (1 - \delta)n_i \rfloor, \quad D_i = m_i O_i.$$

Then $0 \leq m_i < n_i$. Constants belong to $L(D_i)$, so the code is nonzero, and evaluation is injective. Its parameters $[n_i, r_i, d_i]_q$ satisfy

$$r_i \geq m_i + 1 - g_i, \quad d_i \geq n_i - m_i.$$

In particular,

$$\frac{d_i}{n_i} \geq \delta, \quad \frac{r_i}{n_i} \geq \frac{m_i}{n_i} + \frac{1}{n_i} - \frac{g_i}{n_i}.$$

The integer part introduces an error less than $1/n_i$, so $m_i/n_i \rightarrow 1 - \delta$. Also,

$$\frac{g_i}{n_i} = \frac{g_i}{N_i} \frac{N_i}{N_i - 1} \rightarrow \frac{1}{A}.$$

Therefore

$$\liminf_i \frac{r_i}{n_i} \geq 1 - \delta - \frac{1}{A}.$$

To relate this to α_q , observe that

$$A_q(n_i, \lfloor \delta n_i \rfloor) \geq q^{r_i}.$$

A limsup over all lengths is at least the limsup along any sequence of lengths tending to infinity. This proves the stated lower bound for $0 < \delta < 1$. The bound $\alpha_q(\delta) \geq 0$ is automatic. At $\delta = 0$, use $\alpha_q(0) = 1$; at $\delta = 1$, the displayed nontrivial expression is negative, so the bound is again automatic. $\square$

Remark 3.3. We used the Riemann–Roch *inequality* for the dimensions. Imposing $m_i > 2g_i - 2$ would unnecessarily restrict the relative distances for which the argument works. Also, d_i is the actual distance: we only need $d_i \geq n_i - m_i$, not equality. This spells out the integer choices and limits in the asymptotic discussion of [W, Chapter 7].

By choosing a sequence approaching the limsup in (10), the proposition yields, whenever $A(q) > 0$,

$$\alpha_q(\delta) \geq 1 - \delta - \frac{1}{A(q)}. \quad (12)$$

The trivial lower bound zero can always be added. In particular, if a family has limiting ratio $A > 1$, any choice $0 < \delta < 1 - 1/A$ gives both positive asymptotic rate and positive relative minimum distance.

4 Point-counting theorems and the TVZ bound

We now state the geometric results that determine the quality of the preceding construction. The statements follow Walker’s Theorems 7.2 and 7.5–7.7. The point-bound proof sketches appear later; the optimal curve-family construction is an input here.

Theorem 4.1 (Hasse–Weil). *Let $X/\mathbb{F}_q$ be a smooth projective geometrically integral curve of genus g , with $N = \#X(\mathbb{F}_q)$. Then*

$$|N - (q + 1)| \leq 2g\sqrt{q}. \quad (13)$$

Dividing the upper bound by g and letting $g \rightarrow \infty$ gives $A(q) \leq 2\sqrt{q}$. A stronger theorem applies to the asymptotic problem.

Theorem 4.2 (Drinfeld–Vlăduț). *For every prime power q ,*

$$A(q) \leq \sqrt{q} - 1.$$

The Hasse–Weil theorem bounds each individual curve. The Drinfeld–Vlăduț theorem gives a sharper restriction on a sequence whose genus tends to infinity. One exceptionally good curve is not by itself such a sequence.

Theorem 4.3 (Tsfasman–Vlăduț–Zink and Ihara). *If $q = p^{2e}$ is a square prime power, there is a sequence of smooth projective geometrically integral curves $X_i/\mathbb{F}_q$ with $g_i \rightarrow \infty$ and*

$$\lim_{i \rightarrow \infty} \frac{\#X_i(\mathbb{F}_q)}{g_i} = \sqrt{q} - 1.$$

Consequently, for square prime powers,

$$A(q) = \sqrt{q} - 1. \quad (14)$$

Walker attributes the existence result to Tsfasman, Vlăduț, and Zink in the cases $e = 1, 2$, and to Ihara in general. His Theorem 7.6 uses modular curves. The numerical conclusion is what we need today; no tower or modular-curve construction is proved in this lecture.

Corollary 4.4 (The TVZ coding bound). *For a square prime power q and $0 \leq \delta \leq 1$,*

$$\alpha_q(\delta) \geq 1 - \delta - \frac{1}{\sqrt{q} - 1}. \quad (15)$$

Proof. Apply Proposition 3.2 to the family in Theorem 4.3. Alternatively, substitute (14) into (12). $\square$

For $q = 49$ this lower bound is $R \geq 5/6 - \delta$; for $q = 64$ it is $R \geq 6/7 - \delta$. These are lower bounds for the best achievable rate, not restrictions obeyed in that direction by every code. They are useful only where their right-hand side is positive, and are strongest relative to GV in an intermediate range of distances.

5 Comparing the TVZ line with the GV curve

Recall the q -ary entropy function, with $0 \log 0 = 0$:

$$H_q(x) = x \log_q(q-1) - x \log_q x - (1-x) \log_q(1-x).$$

The asymptotic Gilbert–Varshamov bound provides the benchmark

$$\alpha_q(\delta) \geq R_{\text{GV}}(\delta) := 1 - H_q(\delta) \quad \left(0 \leq \delta \leq \theta := 1 - \frac{1}{q}\right).$$

Its proof requires the Hamming-ball asymptotics in Walker’s Lemma 2.8, in addition to the finite GV bound. We use the asymptotic statement here for comparison; we do not treat it as a theorem already proved in the first lecture. The Plotkin bound, proved in Lecture 2, gives

$$\alpha_q(\delta) \leq R_{\text{P}}(\delta) := 1 - \frac{\delta}{\theta} \quad (0 \leq \delta \leq \theta), \quad \alpha_q(\delta) = 0 \quad (\theta \leq \delta \leq 1).$$

For square q , set

$$R_{\text{TVZ}}(\delta) = 1 - \delta - \frac{1}{\sqrt{q} - 1}.$$

Thus the two lower bounds can be combined as

$$\max\{0, R_{\text{GV}}(\delta), R_{\text{TVZ}}(\delta)\} \leq \alpha_q(\delta) \leq R_{\text{P}}(\delta) \quad (16)$$

for $0 \leq \delta \leq \theta$. Beating GV means improving a known lower bound, not contradicting an upper bound or claiming to know the exact function α_q .

5.1 The calculation behind the graphs

The signed difference is

$$\Delta_q(\delta) = R_{\text{TVZ}}(\delta) - R_{\text{GV}}(\delta) = H_q(\delta) - \delta - \frac{1}{\sqrt{q} - 1}.$$

Differentiation on $(0, \theta)$ gives

$$H'_q(\delta) = \log_q \frac{(q-1)(1-\delta)}{\delta}, \quad H''_q(\delta) = -\frac{1}{(\log q)\delta(1-\delta)} < 0.$$

Consequently Δ_q is strictly concave, with a unique maximum at

$$\delta_* = \frac{q-1}{2q-1}, \quad \Delta_q(\delta_*) = \log_q \left(2 - \frac{1}{q}\right) - \frac{1}{\sqrt{q}-1}. \quad (17)$$

To check the value, use $1 - \delta_* = q/(2q - 1)$. Substitution gives

$$H_q(\delta_*) - \delta_* = \log_q(2q - 1) - 1 = \log_q(2 - q^{-1}).$$

The endpoint values are negative:

$$\Delta_q(0) = -\frac{1}{\sqrt{q} - 1}, \quad \Delta_q(\theta) = \frac{1}{q} - \frac{1}{\sqrt{q} - 1} < 0.$$

Whenever the maximum is positive there are therefore exactly two crossings, and TVZ is strictly larger between them.

For completeness, the positivity for every square $q = s^2 \geq 49$ can be checked without numerical plotting. Equation (17) is positive exactly when

$$\frac{s - 1}{2 \log s} \log(2 - s^{-2}) > 1.$$

Both factors $(s - 1)/\log s$ and $\log(2 - s^{-2})$ are positive and increasing for $s > 1$. For the first factor,

$$\frac{d}{ds} \frac{s - 1}{\log s} = \frac{\log s - 1 + 1/s}{(\log s)^2} > 0,$$

since the numerator is zero at $s = 1$ and has derivative $(s - 1)/s^2 > 0$. It is thus enough to check $s = 7$. That inequality is equivalent to

$$97^3 > 7^7, \quad 912673 > 823543.$$

This verifies the comparison asserted after Walker's Theorem 7.7. The derivation above supplies the calculation, rather than relying on the graphs as a proof.

The numerical crossings and maximum differences for the requested alphabets are:

q	Left crossing	Right crossing	Maximum rate gain
49	0.36485356	0.62507424	0.00880153
64	0.28611014	0.70650389	0.02192364

The displayed endpoints are numerical approximations. The exact interval is defined by the two solutions of $\Delta_q(\delta) = 0$. The next two pages show both the full bounds and a magnified view of their difference. The plots are computed directly from the displayed formulas.

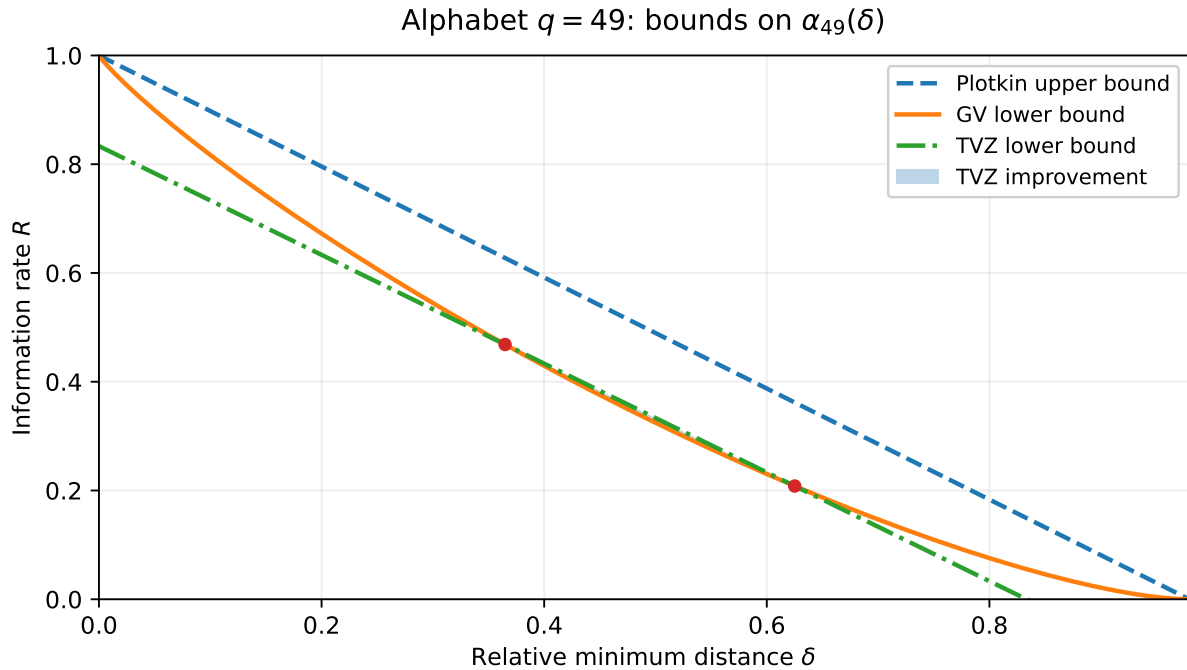


Figure 1: For $q = 49$, the Plotkin line is an upper bound, while GV and TVZ are lower bounds. The shaded region is the improvement supplied by TVZ. Only nonnegative rates are displayed.

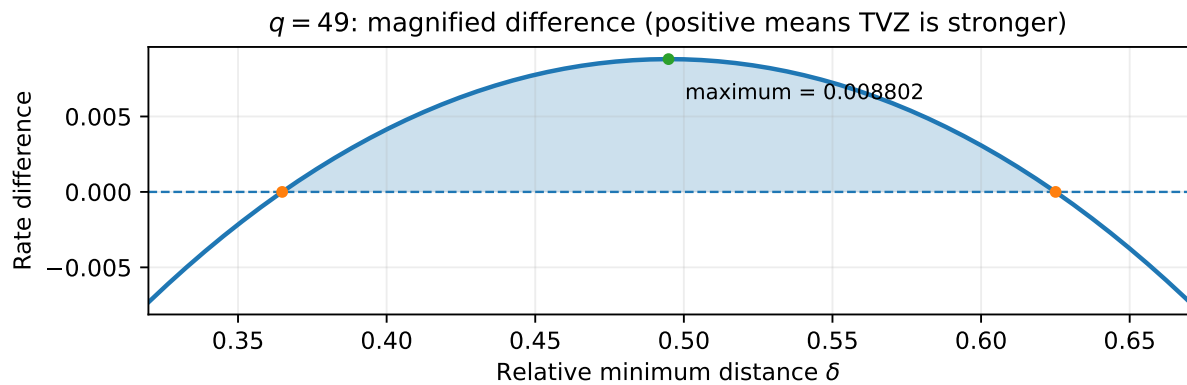


Figure 2: The difference for $q = 49$. Its positive portion lies between the two crossings near 0.36485 and 0.62507. The small vertical scale explains why the improvement is less obvious in the full plot.

At $\delta = 1/2$, for instance, the TVZ rate is $1/3$, while the GV rate is approximately 0.32454546. Thus the improvement is already visible at a simple chosen distance, not only at the maximizing value $\delta_* = 48/97$.

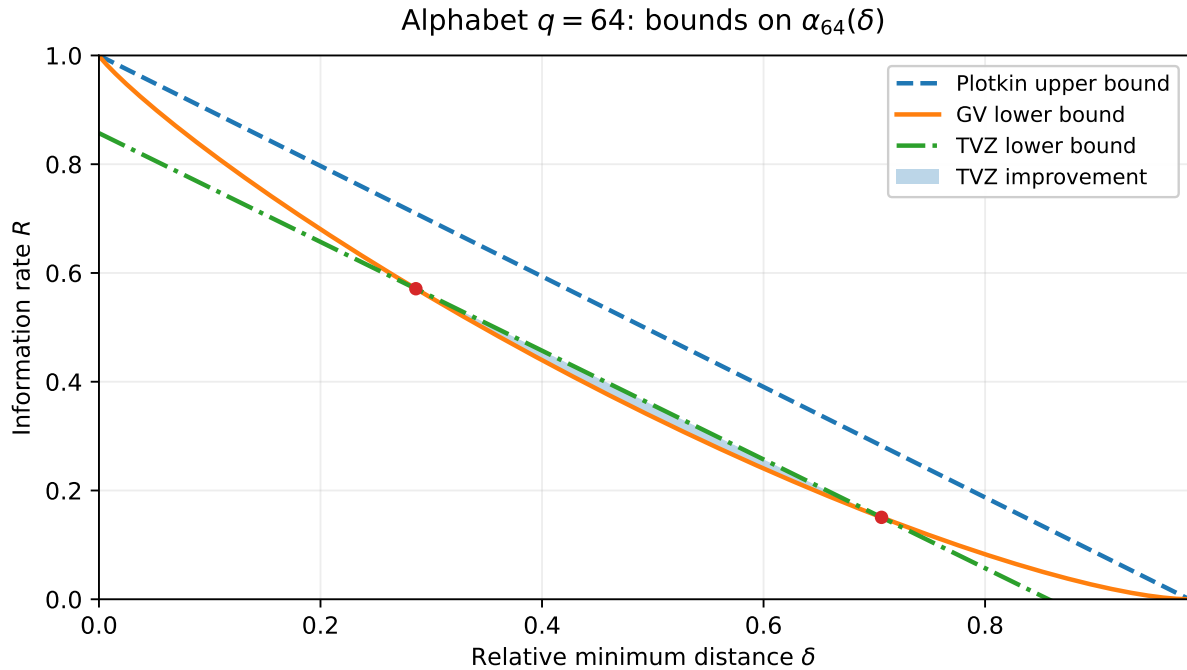


Figure 3: For $q = 64$, TVZ improves GV over a wider interval. The upper Plotkin line still leaves a gap: these estimates do not determine the exact optimal asymptotic rate.

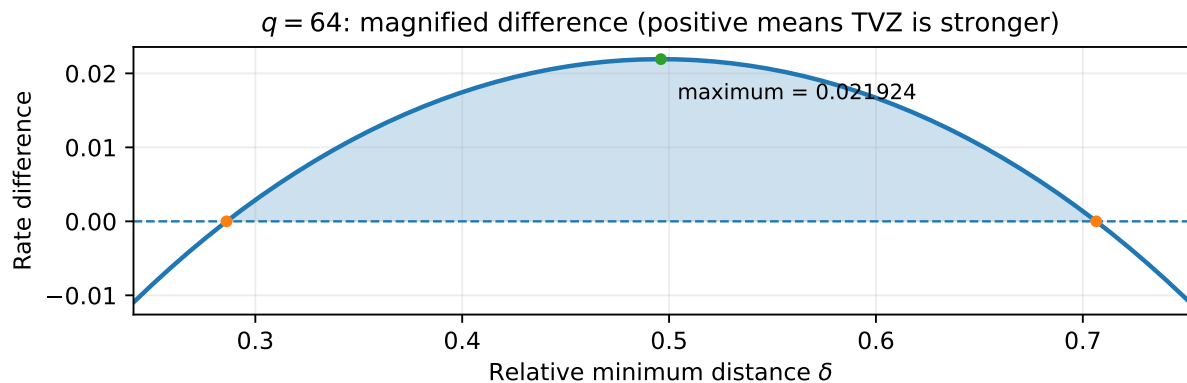


Figure 4: The difference for $q = 64$. TVZ is stronger between the crossings near 0.28611 and 0.70650, and its maximum vertical advantage is about 0.02192.

At $\delta = 1/2$, TVZ gives $R \geq 5/14 \approx 0.35714286$, whereas the GV rate is approximately 0.33522667. The larger improvement is consistent with the greater point-to-genus ratio $A(64) = 7$, compared with $A(49) = 6$.

6 A geometric explanation of Riemann–Roch

We have used Riemann–Roch to construct codes. We now sketch why its formula has the shape it does. This section gives the cohomological explanation over $\mathbb{C}$, then indicates the same algebraic structure over an arbitrary field. The sheaf-cohomology facts and Serre duality are stated inputs to the sketch, rather than results proved from elementary complex analysis. For the general-field statements, see [S]; this explanation is additional to Walker’s statement and Morandi’s omitted proof of Riemann–Roch.

6.1 Local function spaces, cohomology, and genus

Let $X/\mathbb{C}$ be a smooth projective curve, viewed analytically as a compact connected Riemann surface. Its rational functions are global meromorphic functions. If

$$D = \sum_P n_P P,$$

then $\mathcal{O}_X(D)$ denotes the sheaf of meromorphic functions satisfying the bounds prescribed by D locally. In a local coordinate z vanishing at P , its sections have the form

$$z^{-n_P} h(z), \quad h \text{ holomorphic.}$$

A sheaf keeps track of these local spaces and their restriction and gluing rules. This particular sheaf is a line bundle: locally it is generated by the single symbol z^{-n_P} . Its global sections are exactly

$$H^0(X, \mathcal{O}_X(D)) = L(D). \quad (18)$$

Remark 6.1 (Notation). We write $\mathcal{O}_X(D)$, not $\mathcal{O}_D$. The former is the divisor line bundle on the whole curve. When D is effective, $\mathcal{O}_D$ usually means the structure sheaf of the finite subscheme D , which is a different object.

The group $H^1(X, \mathcal{O}_X(D))$ measures a gluing obstruction. Local meromorphic choices may differ on overlaps by sections of $\mathcal{O}_X(D)$. They come from one global choice precisely when those discrepancies can be removed by local corrections. Cohomology identifies equivalent discrepancies and records the remaining obstruction classes. In the principal-parts language, it is local data modulo globally realizable data and permitted local errors.

Write $h^i(D) = \dim_{\mathbb{C}} H^i(X, \mathcal{O}_X(D))$. These spaces are finite-dimensional, and higher coherent cohomology vanishes on a curve. Its Euler characteristic is therefore

$$\chi(X, \mathcal{O}_X(D)) = h^0(D) - h^1(D).$$

Global holomorphic functions on a compact connected Riemann surface are constant, so $h^0(0) = 1$. The genus is

$$\boxed{g = h^1(X, \mathcal{O}_X) = 1 - \chi(X, \mathcal{O}_X).} \quad (19)$$

Over $\mathbb{C}$ it also equals the number of handles of the surface; the topological Euler characteristic is $2 - 2g$, whereas the holomorphic Euler characteristic $\chi(\mathcal{O}_X)$ is $1 - g$. These should not be confused. For a smooth projective geometrically integral curve over any field k , formula (19), with dimensions over k , is the usual algebraic definition. In this smooth setting arithmetic and geometric genus agree.

6.2 Adding a point increases Euler characteristic by one

Let the coefficient of P in D be a . The local spaces for D and $D + P$ are

$$z^{-a}\mathcal{O}_P \subset z^{-a-1}\mathcal{O}_P.$$

The quotient is one-dimensional: it retains just the coefficient of z^{-a-1} . Away from P , nothing changes. This gives the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(D) \longrightarrow \mathcal{O}_X(D + P) \longrightarrow Q_P \longrightarrow 0, \quad (20)$$

where Q_P is a one-dimensional skyscraper sheaf at P . After choosing a local generator, it can be identified with $\mathbb{C}_P$.

The associated long exact sequence of cohomology, written across two lines, is

$$\begin{aligned} 0 \longrightarrow H^0(D) \longrightarrow H^0(D + P) \longrightarrow \mathbb{C} \\ \xrightarrow{\partial} H^1(D) \longrightarrow H^1(D + P) \longrightarrow 0. \end{aligned}$$

Here $H^i(D)$ abbreviates $H^i(X, \mathcal{O}_X(D))$. There are no further terms because Q_P has no higher cohomology and a curve has no coherent cohomology in degree two.

In a finite exact sequence of finite-dimensional vector spaces, the alternating sum of dimensions is zero: this follows by applying rank–nullity at each map. Thus

$$h^0(D) - h^0(D + P) + 1 - h^1(D) + h^1(D + P) = 0,$$

or

$$\chi(\mathcal{O}_X(D + P)) = \chi(\mathcal{O}_X(D)) + 1.$$

Starting with $D = 0$ and adding or subtracting points proves

$$\boxed{h^0(D) - h^1(D) = \deg D + 1 - g.} \quad (21)$$

This is the Euler-characteristic calculation in Riemann–Roch.

Remark 6.2. The map on sheaves in (20) is surjective, but the map $H^0(D + P) \rightarrow \mathbb{C}$ need not be. One new coefficient is available *locally*; the connecting map ∂ records the obstruction to realizing it globally. Thus adding a point does not always increase $\ell(D)$ by one.

6.3 Serre duality identifies the correction term

A meromorphic 1-form is locally written $\eta = h(z) dz$. Its order at P is the order of h ; a change of local coordinate multiplies dz by a holomorphic unit, so the order is well-defined. Let $\mathcal{K}_X$ denote the holomorphic cotangent line bundle. If η_0 is a nonzero global meromorphic 1-form, its zero-and-pole divisor

$$K = (\eta_0) = \sum_P \text{ord}_P(\eta_0) P$$

is a *canonical divisor*.

Every meromorphic 1-form is $f\eta_0$ for a global meromorphic function f . The identity

$$(f\eta_0) = (f) + K$$

shows that

$$H^0(X, \mathcal{K}_X \otimes \mathcal{O}_X(-D)) = \{\eta : (\eta) \geq D\} \cup \{0\} \simeq L(K - D). \quad (22)$$

Replacing η_0 by $u\eta_0$ changes K by the principal divisor (u) ; the canonical divisor class is independent of the choice.

The second input is *Serre duality*:

$$\boxed{H^1(X, \mathcal{O}_X(D))^* \simeq H^0(X, \mathcal{K}_X \otimes \mathcal{O}_X(-D))}. \quad (23)$$

It is the *dual* of H^1 , not H^1 itself, that is naturally a space of meromorphic 1-forms with these order conditions. Combining (21), (22), and (23) yields

$$\boxed{\ell(D) - \ell(K - D) = \deg D + 1 - g}. \quad (24)$$

There is a concrete reason differential forms test gluing obstructions. Given a finite collection of local meromorphic data (a_P) , a form η gives the linear test

$$(a_P) \mapsto \sum_P \text{Res}_P(a_P \eta).$$

If the data come from one global meromorphic function, the global residue theorem makes this sum zero. If $(\eta) \geq D$, the test also ignores the errors allowed by $\mathcal{O}_X(D)$. Serre duality asserts that these tests detect all remaining obstruction classes. This is the complex-geometric motivation for dual principal-part spaces and Weil differentials; it is not a replacement proof of duality.

6.4 The large-degree formula and the general field

Taking $D = 0$ in (24), using $\ell(0) = 1$, gives $\ell(K) = g$. Taking $D = K$ next gives

$$g - 1 = \deg K + 1 - g, \quad \boxed{\deg K = 2g - 2}.$$

For $\deg D > 2g - 2$, the divisor $K - D$ has negative degree. Hence $L(K - D) = 0$, or equivalently $H^1(X, \mathcal{O}_X(D)) = 0$, and

$$\ell(D) = \deg D + 1 - g.$$

For every D , the nonnegativity of $h^1(D)$ in (21) gives the general inequality $\ell(D) \geq \deg D + 1 - g$. These are exactly the statements used in constructing the codes.

Example 6.3. On $\mathbb{P}^1$, take coordinate t and $u = 1/t$ near infinity. Since $dt = -u^{-2}du$, the canonical divisor (dt) is -2∞ . For $D = m\infty$ with $m \geq 0$, the correction space is $L(-(m+2)\infty) = 0$, and Riemann–Roch recovers $\ell(m\infty) = m + 1$.

Over a general field k , one uses algebraic sheaves and algebraic differential forms. At a closed point P , the quotient Q_P in (20) is one-dimensional over $\kappa(P)$, hence has k -dimension $\deg P$. Therefore

$$\chi(\mathcal{O}_X(D + P)) - \chi(\mathcal{O}_X(D)) = \deg P.$$

Adding and subtracting closed points gives the same Euler-characteristic formula with weighted divisor degree. Algebraic Serre duality for a smooth curve identifies the dual correction space with sections of $\Omega_{X/k}^1 \otimes \mathcal{O}_X(-D)$. Thus the same proof structure works over $\mathbb{F}_q$ and over any field; it does not require placing that field inside $\mathbb{C}$. The general duality theorem is an additional input, not something deduced merely by analogy with complex analysis.

7 Hasse–Weil from intersections on a surface

We now sketch an independent geometric proof of Theorem 4.1. Its main input is the *Hodge index theorem* for surfaces, valid also in positive characteristic. This route is not the zeta-function proof in Morandi’s Chapter 7. A reference for the intersection formulation is Hallouin–Perret [HP, Theorem 1, Lemma 3, and §2.5.1].

7.1 Rational points as fixed points of Frobenius

Set $\bar{k} = \bar{\mathbb{F}}_q$, $Y = X_{\bar{k}}$, and $S = Y \times Y$. Let $\varphi : Y \rightarrow Y$ be the base change of the q -power Frobenius endomorphism of X . On geometric projective coordinates,

$$[a_0 : \cdots : a_j] \longmapsto [a_0^q : \cdots : a_j^q].$$

It is a morphism of degree q , its differential is zero, and its fixed points are precisely $X(\mathbb{F}_q)$. The degree is an algebraic multiplicity: Frobenius is nevertheless bijective on geometric points.

In S , consider the diagonal and graph

$$\Delta = \{(P, P) : P \in Y\}, \quad \Gamma = \{(P, \varphi(P)) : P \in Y\}.$$

Their intersection consists of the Frobenius fixed points. At such a point, the tangent directions are (v, v) on the diagonal and $(v, 0)$ on the graph, because $d\varphi = 0$. The two tangent lines are distinct, so each intersection is transverse and has multiplicity one. Consequently

$$\boxed{\Delta \cdot \Gamma = N := \#X(\mathbb{F}_q).} \tag{25}$$

On a surface, divisors are integer combinations of curves. Their intersection number is a symmetric bilinear pairing; for two distinct curves without a common component, it counts intersections with multiplicity. The theory also defines self-intersections, which are *not* the number of points in a curve’s intersection with itself.

7.2 The intersection table

Choose any geometric point $P_0 \in Y(\bar{k})$ and put

$$H = Y \times \{P_0\}, \quad V = \{P_0\} \times Y.$$

These are horizontal and vertical fibers. No $\mathbb{F}_q$ -rational point is needed for this choice. We have

$$\begin{aligned} H^2 = V^2 = 0, \quad H \cdot V = 1, \\ \Delta \cdot H = \Delta \cdot V = 1, \quad \Gamma \cdot H = q, \quad \Gamma \cdot V = 1. \end{aligned}$$

For the graph, the first projection is an isomorphism and the second is φ , explaining the two degrees. The intersection $\Gamma \cdot H$ counts the multiplicity of a fiber, not q distinct geometric points.

The self-intersections are

$$\Delta^2 = 2 - 2g, \quad \Gamma^2 = q(2 - 2g). \tag{26}$$

One way to see this is the adjunction identity for a smooth curve Z on a smooth surface:

$$(K_S + Z) \cdot Z = 2g(Z) - 2.$$

For the product surface, $K_S = \text{pr}_1^* K_Y + \text{pr}_2^* K_Y$. Both Δ and Γ are isomorphic to Y , while the projection degrees give

$$K_S \cdot \Delta = 2(2g - 2), \quad K_S \cdot \Gamma = (1 + q)(2g - 2).$$

Subtracting these numbers in adjunction proves (26). Here we used $\deg K_Y = 2g - 2$, the consequence of Riemann–Roch just explained. Adjunction is included among the surface-theoretic inputs to this sketch.

The entire calculation is summarized by

$\cdot$	H	V	Δ	Γ
H	0	1	1	q
V	1	0	1	1
Δ	1	1	$2 - 2g$	N
Γ	q	1	N	$q(2 - 2g)$

The only unknown in the table is the rational-point count N .

7.3 Project away from the fibers and use Hodge index

Two divisors are numerically equivalent when they have the same intersection with every curve. We work in the real vector space of divisor classes modulo this equivalence. Hodge index says that the intersection pairing is negative definite on the orthogonal complement of an ample divisor. A divisor is ample when a sufficiently high multiple gives a projective embedding. Since $H + V$ is ample on the product, the subspace orthogonal to both H and V has a positive-definite inner product

$$\langle U, W \rangle = -U \cdot W.$$

The projections of Δ and Γ onto this subspace are

$$D_0 = \Delta - H - V, \quad D_1 = \Gamma - H - qV.$$

Indeed, both corrected classes have intersection zero with both fibers. The coefficients can be recovered from the general formula

$$Z^\perp = Z - (Z \cdot V)H - (Z \cdot H)V.$$

Expanding the intersection table gives

$$\begin{aligned} D_0^2 &= (2 - 2g) - 2(1 + 1) + 2 = -2g, \\ D_1^2 &= q(2 - 2g) - 2(q + q) + 2q = -2gq, \\ D_0 \cdot D_1 &= N - (q + 1). \end{aligned}$$

Cauchy–Schwarz for $\langle \cdot, \cdot \rangle$ now implies

$$(N - q - 1)^2 \leq (-D_0^2)(-D_1^2) = (2g)(2gq).$$

Taking square roots proves

$$|N - (q + 1)| \leq 2g\sqrt{q}.$$

The genus-zero case is included: the corrected classes have zero norm, so the mixed pairing is also zero and $N = q + 1$.

The main term $q + 1$ is precisely what was removed in the two fiber directions. The factor $2g$ comes from the corrected diagonal's self-intersection, and $\sqrt{q}$ comes from comparing it with the Frobenius graph by Cauchy–Schwarz. Replacing φ by φ^r repeats the same proof with degree q^r and gives

$$|\#X(\mathbb{F}_{q^r}) - (q^r + 1)| \leq 2gq^{r/2}.$$

This argument uses the characteristic- p Hodge index theorem; it does not assume that Frobenius lifts to a map of complex curves.

8 Hasse–Weil from the zeta function and function-field RH

The second route follows Morandi's Chapter 7. It organizes divisor counts into a generating function, states the structure theorem supplied by Riemann–Roch, and then applies Weil's Riemann hypothesis. In this route the root-location theorem is a stated deep input, just as Hodge index was the stated input to the preceding sketch.

8.1 Effective divisors and the zeta function

Let $F = \mathbb{F}_q(X)$, with exact constant field $\mathbb{F}_q$. The norm of a closed point is

$$N(P) = \#\kappa(P) = q^{\deg P}.$$

For an effective divisor $E = \sum_P e_P P$, its norm is

$$N(E) = \prod_P N(P)^{e_P} = q^{\deg E}.$$

The zeta function is initially defined for $\operatorname{Re}(s) > 1$ by

$$\zeta_F(s) = \sum_{E \geq 0} q^{-s \deg E}. \quad (27)$$

Let

$$A_n = \#\{E \geq 0 : \deg E = n\}, \quad Z_F(t) = \sum_{n=0}^{\infty} A_n t^n.$$

Then

$$\boxed{\zeta_F(s) = Z_F(q^{-s})}. \quad (28)$$

There are only finitely many effective divisors of any fixed degree. A divisor of degree n can involve only closed points of degrees at most n , of which there are finitely many, and their coefficients are bounded. The power series converges for $|t| < q^{-1}$; its rational continuation is described below.

Remark 8.1 (Include the zero divisor). The sums include $E = 0$, so $A_0 = 1$. Summing only over nonzero effective divisors gives $\zeta_F(s) - 1$, not the normalization used in the structure theorem or in RH. Morandi's initial series displays on p. 67 start at $n = 1$, but his Corollary 7.6 and coefficient calculation in Theorem 7.18 use $n = 0$ and $A_0 = 1$. We keep this normalization throughout.

An effective divisor of degree one is exactly one rational point. Therefore

$$A_1 = N = \#X(\mathbb{F}_q), \quad Z_F(t) = 1 + Nt + \cdots.$$

For $n > 1$, A_n is not the number of $\mathbb{F}_{q^n}$ -points. For example, a degree-two divisor can be the sum of two rational points, a doubled rational point, or one degree-two closed point. It is a divisor count, not a point count.

Unique expression of an effective divisor as a sum of closed points also gives the Euler product

$$Z_F(t) = \prod_{P \text{ closed}} \frac{1}{1 - t^{\deg P}}.$$

The factor indexed by P is $1 + t^{\deg P} + t^{2\deg P} + \dots$, allowing every nonnegative multiplicity at P . These definitions and the product are Morandi's §7.3 and Proposition 7.10.

8.2 The polynomial numerator and its first coefficient

Riemann–Roch enters through the number of effective divisors in a fixed divisor class. Such divisors have the form $D + (f)$ with $0 \neq f \in L(D)$. Two functions determine the same divisor precisely when their ratio is a nonzero constant. Consequently

$$\#\{E \geq 0 : E \sim D\} = \frac{q^{\ell(D)} - 1}{q - 1}. \quad (29)$$

For large degree, $\ell(D) = \deg D + 1 - g$, so these counts involve geometric progressions. Organizing the divisor classes and summing the tails yields rationality. The symmetry between D and $K - D$ in full Riemann–Roch supplies a functional equation. We state the resulting theorem; the divisor-class and constant-field arguments are developed in Morandi's Chapter 7, rather than repeated in this sketch.

Theorem 8.2 (Structure of the zeta function; Morandi, Theorem 7.18). *There is a polynomial $L_F(t) \in \mathbb{Z}[t]$ such that*

$$Z_F(t) = \frac{L_F(t)}{(1-t)(1-qt)}. \quad (30)$$

It has degree $2g$, constant coefficient 1, leading coefficient q^g , and satisfies

$$L_F(t) = q^g t^{2g} L_F\left(\frac{1}{qt}\right).$$

The rational function $Z_F(t)$ has simple poles at $t = 1$ and $t = 1/q$. Its numerator coefficient of t is $N - (q + 1)$. For $g = 0$ the polynomial is $L_F(t) = 1$, and the coefficient assertion says $N = q + 1$.

Following Morandi, $L_F(t)$ is called the *L-polynomial*. The subscript distinguishes it from the vector space $L(D)$. The coefficient calculation is particularly simple:

$$\begin{aligned} L_F(t) &= (1-t)(1-qt)Z_F(t) \\ &= (1 - (q+1)t + qt^2)(1 + Nt + \dots) \\ &= 1 + (N - q - 1)t + \dots \end{aligned}$$

Thus the error in the rational-point count is already encoded in the linear coefficient of a degree- $2g$ polynomial.

Over $\mathbb{C}$, write its factorization as

$$L_F(t) = \prod_{j=1}^{2g} (1 - \alpha_j t). \quad (31)$$

The α_j are the *reciprocal roots*: the roots themselves are $\beta_j = \alpha_j^{-1}$. Comparing linear coefficients in (31) gives Morandi's Corollary 7.19:

$$N = q + 1 - \sum_{j=1}^{2g} \alpha_j. \quad (32)$$

No root-size estimate has been used to obtain this identity.

8.3 Weil's Riemann hypothesis and the point bound

Theorem 8.3 (Riemann hypothesis for curves over finite fields; Weil). *The zeros of the meromorphic continuation of $\zeta_F(s)$ lie on the line*

$$\operatorname{Re}(s) = \frac{1}{2}.$$

Equivalently, every reciprocal root in (31) satisfies

$$|\alpha_j| = \sqrt{q}. \quad (33)$$

This is a proved theorem, not an unproved hypothesis in the function-field setting. It is Morandi's Theorem 7.20, which he states without proof. Rationality and the functional equation do not by themselves establish it.

To see the equivalence, use $t = q^{-s}$. Writing $s = \sigma + i\tau$ gives

$$q^{-s} = e^{-\sigma \log q} e^{-i\tau \log q}, \quad |q^{-s}| = q^{-\sigma}.$$

The roots of the numerator are the zeros of the rationally continued $Z_F(t)$; the two displayed denominator poles are not canceled. Thus the critical line $\sigma = 1/2$ corresponds to the circle $|t| = q^{-1/2}$. In particular,

$$|\beta_j| = q^{-1/2} \quad \text{for roots,} \quad |\alpha_j| = q^{1/2} \quad \text{for reciprocal roots.} \quad (34)$$

The absolute value here is the ordinary complex absolute value. The original curve is still defined over $\mathbb{F}_q$; the complex numbers arise by factoring its integer polynomial $L_F(t)$.

Using (32) and (33), the triangle inequality gives

$$|N - (q + 1)| = \left| \sum_{j=1}^{2g} \alpha_j \right| \leq \sum_{j=1}^{2g} |\alpha_j| = 2g\sqrt{q}.$$

This recovers the Hasse–Weil point bound, Morandi's Corollary 7.21.

Example 8.4 (The cubic from Lecture 2). For $X_0 : y^2 = x^3 + 2x + 2$ over $\mathbb{F}_3$, we found $g = 1$ and $N = 1$. The constant, linear, and leading coefficients determine

$$L_F(t) = 1 - 3t + 3t^2.$$

Its reciprocal roots solve $z^2 - 3z + 3 = 0$, so

$$\alpha_1 = \frac{3 + i\sqrt{3}}{2}, \quad \alpha_2 = \frac{3 - i\sqrt{3}}{2}, \quad |\alpha_1| = |\alpha_2| = \sqrt{3}.$$

Their sum is three, and the point formula reads $N = 4 - 3 = 1$. This is a direct illustration of the theorem for our example, not a proof of the general root-location statement.

The two Hasse–Weil approaches have different starting inputs. The surface argument bounds the point count directly by Hodge index. The zeta-function argument first encodes the count in reciprocal roots and then uses Weil's RH to control each root. In the latter route, Riemann–Roch determines the shape of the expression; the root theorem supplies the missing absolute values.

Lecture summary

The main construction and its consequences are

$$L(D) \xrightarrow{\text{ev}_{\mathcal{P}}} \mathbb{F}_q^n, \quad \ker \text{ev}_{\mathcal{P}} = L(D - E), \quad E = P_1 + \cdots + P_n,$$

$$\deg D < n \implies r = \ell(D) \geq \deg D + 1 - g, \quad d \geq n - \deg D,$$

with equality in the dimension formula when $\deg D > 2g - 2$. A growing family over one fixed field with $N_i/g_i \rightarrow A$ yields

$$\alpha_q(\delta) \geq 1 - \delta - \frac{1}{A}.$$

For square q , the TVZ–Ihara existence theorem and Drinfeld–Vlăduț upper bound give $A(q) = \sqrt{q} - 1$. The resulting TVZ line is above GV on a nonempty interval for square $q \geq 49$.

Behind these coding estimates are two kinds of geometry. Riemann–Roch counts global functions from local conditions, with the genus and a duality correction measuring the failure of local choices to be globally realizable. Hasse–Weil controls the supply of rational points, either through intersections with the Frobenius graph or through the complex sizes of the zeta function’s reciprocal roots.

References

- [W] J. L. Walker, *Codes and Curves*. Student Mathematical Library, vol. 7, American Mathematical Society. For this lecture: §2.2 (asymptotic parameters, GV, and Plotkin), Chapter 6 (AG codes), and Chapter 7, especially Theorems 7.2 and 7.5–7.7 (point counts and TVZ).
- [M] P. J. Morandi, *Error Correcting Codes and Algebraic Curves*, Mathematics 601 lecture notes, Fall 2001. Supplied 80-page notes: Chapter 5, especially Theorem 5.2 and Corollary 5.3; Chapter 7, especially Lemma 7.5 and Theorems 7.18, 7.20 and Corollaries 7.19, 7.21. The zero-divisor normalization is explained in Section 8.
- [S] The Stacks Project. Supplementary references for the geometric Riemann–Roch sketch: [Tag 0BY6](#), [The genus of a curve](#); [Tag 0AYQ](#), [Degrees on curves](#); and [Tag 0BS2](#), [Duality for proper curves](#). These provide the general algebraic formulation; they are not prerequisites for the elementary code-parameter proofs.
- [HP] E. Hallouin and M. Perret, *From Hodge Index Theorem to the number of points of curves over finite fields*, [arXiv:1409.2357](#). Supplementary reference for the surface proof: Theorem 1, Lemma 3, and §2.5.1. This proof is separate from Morandi’s zeta-function presentation.